

Comparative Analysis of Inflammatory Markers Between Control and Treatment Groups

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Objective

To evaluate whether a simulated treatment reduces the pro-inflammatory cytokines IL-6 and TNF- α compared to a control group, and to demonstrate a reproducible biomedical-data workflow (dataset → descriptive statistics → inferential test → visualisation → interpretation).

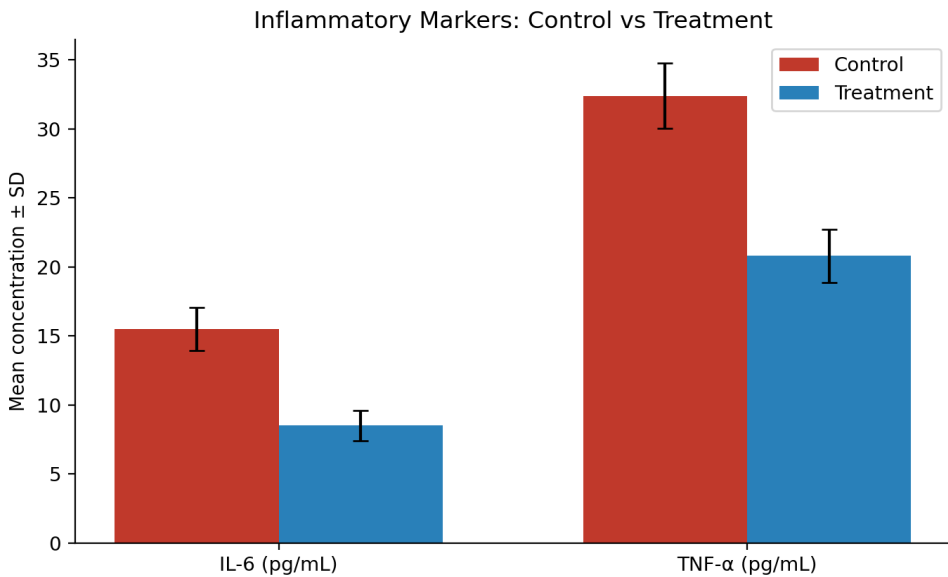
Methods

A simulated dataset of 20 patients (10 Control, 10 Treatment) with measurements of IL-6 and TNF- α (pg/mL) was analysed in Python (pandas, SciPy, matplotlib). Group means and standard deviations were computed. Between-group differences were tested with Welch's two-sample t-test (unequal variances assumed). Significance threshold: $p < 0.05$.

Results

Group	IL-6 (mean $\pm$ SD)	TNF- α (mean $\pm$ SD)	n
Control	15.50 $\pm$ 1.58	32.40 $\pm$ 2.37	10
Treatment	8.50 $\pm$ 1.08	20.80 $\pm$ 1.93	10

Welch's t-test — IL-6: $t = 11.56$, $p = 3.79\text{e-}09$. TNF- α : $t = 12.01$, $p = 8.04\text{e-}10$. Both cytokines were significantly lower in the Treatment group.



Conclusion

The treatment group showed markedly lower mean IL-6 and TNF- α than the control group, consistent with an anti-inflammatory effect. The result is statistically significant for both markers, suggesting the treatment may modulate pro-inflammatory cytokine activity — a finding relevant to immunological and regenerative-medicine contexts. Tools used: Python (pandas, SciPy, matplotlib).